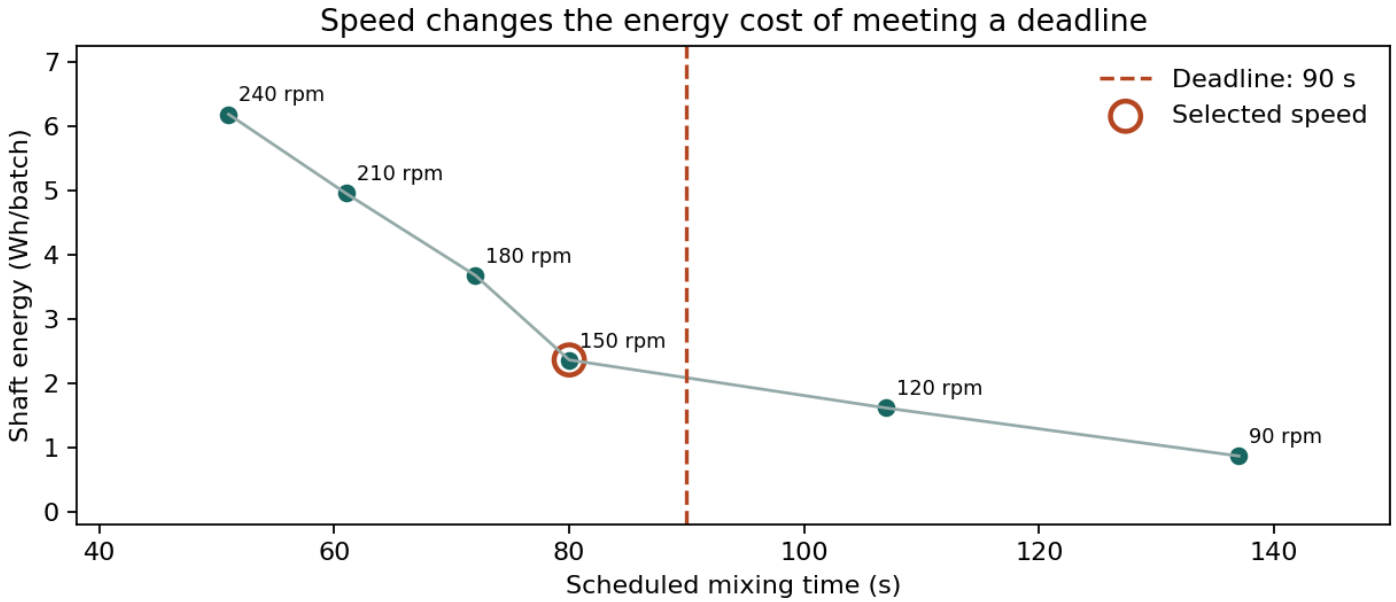
How long should the mixer run?

500 L working volume / One fixed impeller / 90-second production deadline

A soluble dye is blended into a water-like liquid. Six shaft speeds compete on energy per batch. Each speed gets its own stop time, long enough for both virtual probes to settle in every one of its four simulated trials.



150 rpm for 80 seconds

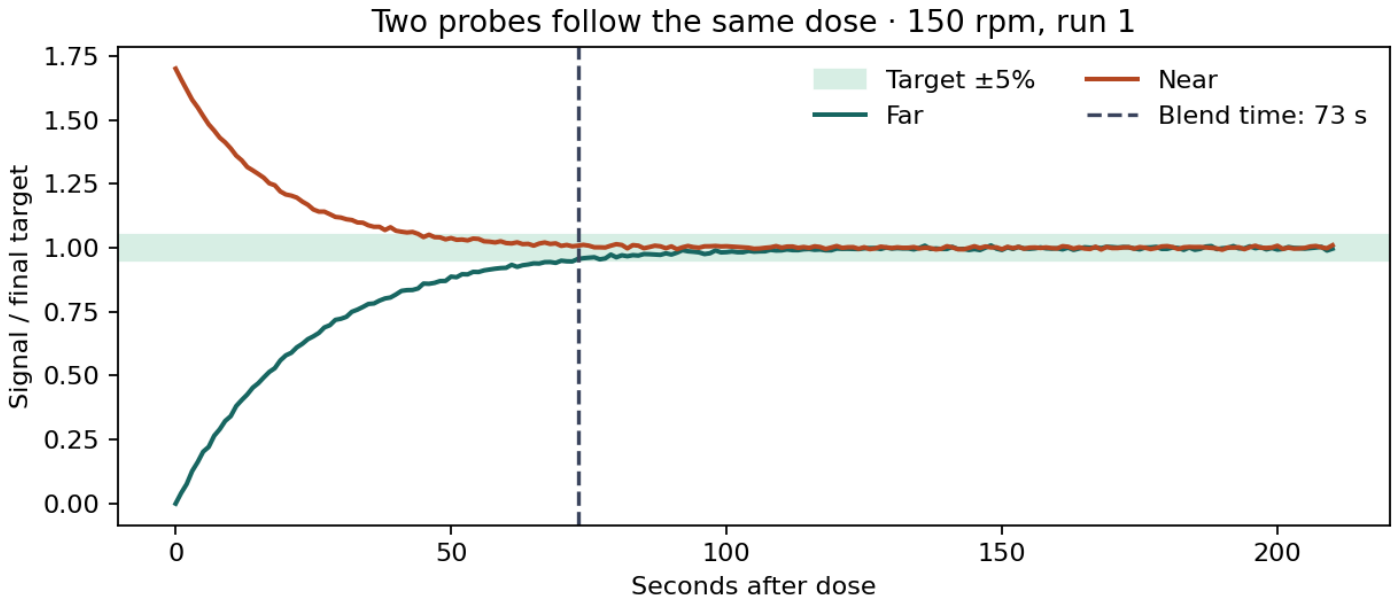
This schedule uses 2.37 Wh of shaft energy per batch. The 240 rpm reference finishes in 51 seconds and uses 6.19 Wh. Slowing down trades 29 seconds for 3.82 Wh of modeled shaft-energy savings per batch (61.7%), with 10 seconds left before the deadline.

Speed (rpm)	Stop time (s)	Power (W)	Energy (Wh)	Meets 90 s?
90	137	23.0	0.88	No
120	107	54.6	1.62	No
150	**80**	**106.7**	**2.37**	**Yes**
180	72	184.4	3.69	Yes
210	61	292.7	4.96	Yes
240	51	437.0	6.19	Yes

The lower-energy 120 rpm option takes 107 seconds. With a relaxed 110-second deadline it becomes the preferred tested speed. A tighter 75-second deadline instead favors 180 rpm for 72 seconds at 3.69 Wh.

From tracer signal to stop time

Two local probes / Four trials per speed / One-second sampling interval



Read the endpoint from both traces

After a simulated dose, the near probe sees a dye-rich plume and the far probe starts below target. Each signal relaxes exponentially toward 1.0, with random run variability and additive noise. Take the first second after the last excursion of either signal outside 0.95-1.05. Require ten further recorded seconds to confirm settling. Use the slowest endpoint across four trials as the scheduled mixing time for that speed.

Convert the timer setting into energy

For one 0.35 m impeller, assume a turbulent power number of 1.3, density of 1,000 kg/m3 and viscosity of 0.001 Pa s. Shaft power is Np times density times speed cubed times diameter to the fifth power. At 150 rpm, N = 2.5 revolutions/s, giving 106.69 W. Multiplying by 80 s and dividing by 3,600 s/h gives 2.37 Wh.

What the model can establish

The choice minimizes modeled shaft energy among the six tested speeds. Four simulated trials cannot establish future reliability. For a plant decision, measure blend times at representative probe locations and verify mixer torque or drive input. Whole-batch electricity use also includes losses and other equipment.

Mixing in Chemical Reactors / University of Michigan
Lectures on Modeling Reacting Flows in Stirred Tanks / Andre Bakker